

NUMERICAL METHODS

Unit-1: Representation of Numbers and Errors

Comprehensive Study Notes



1.1 Introduction to Numerical Methods

Numerical Methods are techniques used to solve mathematical problems that cannot be solved analytically (exactly). They provide **approximate solutions** using arithmetic operations.

Why Numerical Methods?

Reason	Description
Complexity	Many real-world problems have no exact solution
Speed	Computers can perform millions of calculations quickly
Approximation	Acceptable error in many applications
Practicality	Some problems are too complex for analytical methods

Numerical Methods vs Analytical Methods

Feature	Analytical	Numerical
Solution	Exact	Approximate
Speed	May be slow	Fast with computers
Error	None	Present
Complexity	Limited	Handles complex problems
Application	Simple equations	Real-world problems

1.2 Representation of Numbers

1.2.1 Decimal to Binary Conversion

Computers use the **binary number system** (base-2) to represent data.

Decimal to Binary (Integer Part)

Method: Divide by 2 repeatedly and collect remainders (bottom to top).

Example 1: Convert 25 (decimal) to binary

Division	Quotient	Remainder
-----+-----+-----		
25 / 2	12	1
12 / 2	6	0
6 / 2	3	0
3 / 2	1	1
1 / 2	0	1

Read remainders bottom to top: $25 = 11001_2$

Example 2: Convert 13 (decimal) to binary

$13 / 2 = 6 \text{ R } 1$
 $6 / 2 = 3 \text{ R } 0$
 $3 / 2 = 1 \text{ R } 1$
 $1 / 2 = 0 \text{ R } 1$

Answer: $13 = 1101_2$

Decimal to Binary (Fractional Part)

Method: Multiply fractional part by 2 repeatedly and collect integer parts (top to bottom).

Example: Convert 0.625 (decimal) to binary

Step	Operation	Result	Integer
1	0.625×2	1.25	1
2	0.25×2	0.5	0
3	0.5×2	1.0	1

Answer: $0.625 = 0.101_2$

Complete Example: Convert 25.625 to binary

Integer Part (25): $25 = 11001_2$

Fractional Part (0.625): $0.625 = 0.101_2$

Answer: $25.625 = 11001.101_2$

Binary to Decimal Conversion

Method: Multiply each binary digit by 2 raised to its position power.

Example: Convert 1101_2

$$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 8 + 4 + 0 + 1 = 13$$

Quick Conversion Table

Decimal	Binary	Decimal	Binary
0	0	8	1000
1	1	9	1001
2	10	10	1010
3	11	11	1011
4	100	12	1100

5	101	13	1101
6	110	14	1110
7	111	15	1111

Fractions in Binary

Fraction	Binary	Fraction	Binary
0.5	0.1	0.125	0.001
0.25	0.01	0.0625	0.0001
0.75	0.11	0.375	0.011

1.3 Floating Point Representation of Numbers

Floating point representation stores real numbers in computers with three parts: **Sign, Mantissa, and Exponent.**

$$Number = \pm Mantissa \times Base^{Exponent}$$

$$Example: 0.1011 \times 2^3$$

IEEE 754 Single Precision (32-bit)

Sign (1 bit)	Exponent (8 bits)	Mantissa (23 bits)
0 = Positive 1 = Negative	Bias: 127 E = Actual + 127	Hidden bit: 1.x

IEEE 754 Double Precision (64-bit)

Sign (1 bit)	Exponent (11 bits)	Mantissa (52 bits)
0 = Positive 1 = Negative	Bias: 1023	Hidden bit: 1.x

Example: Convert 25.625 to IEEE 754 Single Precision

Step 1: Convert to binary: $25.625 = 11001.101_2$

Step 2: Normalize: 1.1001101×2^4

Step 3: Sign bit: 0 (positive)

Step 4: Exponent: $4 + 127 = 131 = 10000011_2$

Step 5: Mantissa: 10011010000000000000000 (23 bits)

Final: 0 10000011 10011010000000000000000

Special Values

Value	Sign	Exponent	Mantissa
Zero	0	All 0s	All 0s
Infinity	0/1	All 1s	All 0s
NaN	0/1	All 1s	Non-zero
Denormal	0/1	All 0s	Non-zero

1.4 Integer and Floating Point Arithmetic

Integer Arithmetic

```
Addition:      5 + 3 = 8
Subtraction:    10 - 4 = 6
Multiplication: 6 x 7 = 42
Division:       17 / 5 = 3 (integer division)
Modulo:         17 % 5 = 2 (remainder)
```

Floating Point Arithmetic

Feature	Description
Range	Very large ($\sim 10^{-308}$ to 10^{308})
Precision	Limited (7-16 decimal digits)
Operations	Addition, Subtraction, Multiplication, Division
Errors	Rounding errors occur

Precision Issues - Loss of Significance

$$x = 1.0000001, y = 1.0000000$$
$$x - y = 0.0000001 \text{ (Only 7 digits of precision remain)}$$

1.5 Different Types of Errors

1.5.1 Absolute Error

Definition: The difference between the exact value and the approximate value.

$$E_A = |X - X'|$$

Where: X = Exact value, X' = Approximate value

Exact value: $\pi = 3.141592653589793$

Approximate: $\pi' = 3.1416$

Absolute Error = $|3.141592653589793 - 3.1416| = 0.000007346410207$

1.5.2 Relative Error

$$E_R = |X - X'| / |X|$$

Exact: 100 Approx: 99.5

Absolute Error = 0.5

Relative Error = $0.5/100 = 0.005 = 0.5\%$

1.5.3 Percentage Error

$$E_P = (|X - X'| / |X|) \times 100\%$$

Exact: 50 Approx: 49.8
Percentage Error = $(0.2/50) \times 100 = 0.4\%$

1.5.4 Round-Off Error

Error caused by limiting the number of digits in calculations.

$\pi = 3.141592653589793$
Round to 4 decimals: 3.1416
Round-off error = 0.000007346410207

1.5.5 Truncation Error

Error caused by truncating a series or process before completion.

$$e^x = 1 + x + x^2/2! + x^3/3! + x^4/4! + \dots$$

If truncated after x^3 :

$$e^x \approx 1 + x + x^2/2! + x^3/3!$$

$$\text{Truncation error} = x^4/4! + x^5/5! + \dots$$

1.5.6 Inherent Error

Error present in the input data itself from measurement, data entry, rounding of constants (π , e), or previous calculations.

1.6 Error in the Approximation of a Function

Taylor Series Approximation

$$f(x) = f(a) + f'(a)(x-a) + f''(a)(x-a)^2/2! + f'''(a)(x-a)^3/3! + \dots$$

Truncating after n terms:

$$f(x) \approx \sum_{k=0}^n f^{(k)}(a)(x-a)^k / k!$$

Error Term (Remainder)

$$R_n(x) = f^{(n+1)}(\xi) (x-a)^{n+1} / (n+1)!$$

Where ξ is between a and x .

Example: Approximating e^x at $x = 1$ with 3 terms

Exact $e = 2.718281828$

Approx = $1 + 1 + 0.5 = 2.5$

Error = 0.218281828

Maclaurin Series (Taylor centered at $a = 0$)

$$f(x) = f(0) + f'(0)x + f''(0)x^2/2! + f'''(0)x^3/3! + \dots$$

Common Maclaurin Series

Function	Series
e^x	$1 + x + x^2/2! + x^3/3! + x^4/4! + \dots$
$\sin x$	$x - x^3/3! + x^5/5! - x^7/7! + \dots$
$\cos x$	$1 - x^2/2! + x^4/4! - x^6/6! + \dots$
$\ln(1+x)$	$x - x^2/2 + x^3/3 - x^4/4 + \dots$
$(1+x)^n$	$1 + nx + n(n-1)x^2/2! + \dots$

1.7 Error in Series Approximation

Alternating Series Estimation Theorem

$$|R_n| \leq |a_{n+1}|$$

Example: Approximating $\sin(0.5)$

Terms	Value	Error Bound
x	0.5	0.020833
$x - x^3/6$	0.479167	0.000260
$x - x^3/6 + x^5/120$	0.479427	0.00000155

Error Propagation Rules

Operation	Error Propagation
Addition/Subtraction	Absolute errors add
Multiplication/Division	Relative errors add
Powers	Error multiplied by power

Example

If $x = 2.0 \pm 0.1$, $y = 3.0 \pm 0.2$

Sum: $x + y = 5.0 \pm 0.3$

Product: $x \times y = 6.0 \pm 0.7$ (relative errors: $0.05 + 0.067 = 0.117$)

1.8 Complete Examples

Example 1: Convert 47.375 to binary

Integer part (47):

$$47 / 2 = 23 \text{ R } 1$$

$$23 / 2 = 11 \text{ R } 1$$

$$11 / 2 = 5 \text{ R } 1$$

$$5 / 2 = 2 \text{ R } 1$$

$$2 / 2 = 1 \text{ R } 0$$

$$1 / 2 = 0 \text{ R } 1$$

$$47 = 101111_2$$

Fractional part (0.375):

$$0.375 \times 2 = 0.75 \rightarrow 0$$

$$0.75 \times 2 = 1.5 \rightarrow 1$$

$$0.5 \times 2 = 1.0 \rightarrow 1$$

$$0.375 = 0.011_2$$

$$\text{Answer: } 47.375 = 101111.011_2$$

Example 2: Convert -12.625 to IEEE 754 Single Precision

Step 1: Binary: $12.625 = 1100.101_2$

Step 2: Normalize: 1.100101×2^3

Step 3: Sign bit: 1 (negative)

Step 4: Exponent: $3 + 127 = 130 = 10000010_2$

Step 5: Mantissa: 1001010000000000000000

$$\text{Answer: } 1 \ 10000010 \ 1001010000000000000000$$

Example 3: Error Calculation

Problem: Approximate π as 3.14. Find absolute, relative, and percentage errors.

$$\text{Exact } \pi = 3.141592653589793$$

Approx $\pi' = 3.14$

Absolute Error = $|3.141592653589793 - 3.14| = 0.001592653589793$

Relative Error = $0.001592653589793 / 3.141592653589793 = 0.0005069$

Percentage Error = $0.0005069 \times 100 = 0.05069\%$

Example 4: Taylor Series Error for $\cos(0.3)$

$$\cos x = 1 - x^2/2! + x^4/4! - x^6/6! + \dots$$

For $x = 0.3$:

Approx = $1 - 0.045 + 0.0003375 = 0.9553375$

Exact $\cos(0.3) = 0.955336489$ (calculator)

Error = $0.955336489 - 0.9553375 = -0.000001011$

Truncation Error $\sim 0.36/6! = 0.0000010125$ (next term)

1.9 Key Formulas Summary

Number Systems

System	Base	Digits
Binary	2	0, 1
Octal	8	0-7
Decimal	10	0-9
Hexadecimal	16	0-9, A-F

Error Formulas

Error Type	Formula
Absolute Error	$E_A = X - X' $
Relative Error	$E_R = E_A / X $
Percentage Error	$E_P = E_R \times 100\%$

Floating Point Characteristics

Type	Bits	Sign	Exponent	Mantissa	Bias
Single	32	1	8	23	127
Double	64	1	11	52	1023

1.10 Previous Year Questions

Short Questions (2 marks)

1. Convert 101101_2 to decimal.
2. Convert 45_{10} to binary.
3. Define absolute error.
4. What is relative error?
5. Define rounding error.
6. What is truncation error?
7. Write the Taylor series for e^x .
8. What is the binary representation of 0.5?

Medium Questions (5 marks)

1. Convert 101.101_2 to decimal.
2. Convert 37.625_{10} to binary.
3. Explain different types of errors in numerical methods.
4. Find the absolute, relative, and percentage error if π is approximated as 3.1416.
5. Explain the floating point representation of numbers.

Long Questions (12 marks)

1. (a) Convert the following decimal numbers to binary: (i) 47.375 (ii) 101.101_2 to decimal.
(b) Explain different types of errors in numerical computation with examples.

2. (a) Explain different types of errors in numerical computation with suitable examples.
(b) Convert 37.625 to IEEE 754 single precision format.
3. (a) Explain Taylor series expansion of a function. Give error term.
(b) Approximate $e^{0.5}$ using 4 terms and find error.

Quick Reference

Concept	Key Point
Binary Conversion (Integer)	Divide by 2, read remainders bottom to top
Binary Conversion (Fraction)	Multiply by 2, read integers top to bottom
IEEE 754 Single Format	1-bit Sign 8-bit Exponent (+127) 23-bit Mantissa
Error Types	Absolute, Relative, Percentage, Round-off, Truncation, Inherent
Taylor Series	$f(x) = \sum f^{(k)}(a)(x-a)^k/k!$